

D374

Market Research

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
-----------------------------	---	------------------------	----------------------------------

Items

1.

A coffee brand wants to know how its new cold brew is perceived. The team reads a published industry report on beverage trends and also runs its own taste test with 200 shoppers. The brand's warehouse is in Seattle, and it just hired a new CMO. Which is primary data?

- A. The taste-test results
- B. Both
- C. The industry report
- D. Neither, because only surveys produce primary data

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A bank wants a survey of its 50,000 customers that reflects the exact mix of its three customer tiers. It divides customers by tier and randomly selects from each tier in proportion to its size. The survey will be sent by email on a Tuesday. Which sampling method is this?

- A. Stratified random sampling
- B. Simple random sampling
- C. Cluster sampling
- D. Quota sampling

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A researcher stops shoppers at one mall entrance until she has 50 men and 50 women. What kind of sample is this?

- A. Simple random sample
- B. Stratified random sample
- C. Quota (non-probability) sample
- D. A probability sample, since both genders are included

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A retailer wants to estimate the share of customers who would join a loyalty program within ± 5 percentage points at 95% confidence ($z = 1.96$), with no prior estimate, so $p = 0.5$. The retailer has 120,000 customers and operates in four states. What minimum sample size is needed?

- A. 196
- B. 385
- C. 384
- D. 12,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Keeping 95% confidence and $p = 0.5$, a researcher cuts the margin of error from ± 5 to ± 2.5 points. What happens to the required sample size?

- A. It roughly quadruples, from 385 to about 1,537.
- B. It halves.
- C. It stays the same, since confidence did not change.
- D. It doubles.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

An A/B test compares a new checkout page with the old one. H_0 is no difference in conversion. The test gives $p = 0.03$ with $\alpha = 0.05$. The new page uses a green button, and the test ran for 14 days. What should the analyst conclude?

- A. Fail to reject H_0 .
- B. Reject H_0 ; the difference is statistically significant.
- C. The green button caused the lift.
- D. There is a 97% chance the new page is better.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

With $\alpha = 0.01$, a test returns $p = 0.04$. What is the conclusion?

- A.** Reject H_0 , because p is below 0.05.
- B.** Reject H_0 .
- C.** Accept H_0 as proven true.
- D.** Fail to reject H_0 .

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A company concludes that a new ad raised sales, when in reality the ad had no effect. The ad starred a famous chef and cost \$2 million. What error is this?

- A.** Sampling error
- B.** Type I error (false positive)
- C.** Type II error (false negative)
- D.** Nonresponse bias

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A test fails to detect a real effect of a price change on demand. Which error is this, and what reduces its chance?

- A.** Type II; lower α further.
- B.** Type I; lower α .
- C.** No error; a non-significant result always means no effect.
- D.** Type II; increase the sample size to raise power.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A survey for a 24-hour gym chain with 60 locations asks: 'How satisfied are you with our friendly staff and clean facilities?' The survey will go to 10,000 members by text. What is the main problem?

- A.** Nothing; positive wording always improves response quality.
- B.** The sample is too small.
- C.** It is double-barreled and leading: it asks about two things and assumes staff are friendly and facilities clean.
- D.** It uses an interval scale.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A survey asks runners to rank five shoe brands from 1 (favorite) to 5 (least favorite). Responses came from 800 runners at a marathon expo. What level of measurement is this?

- A.** Interval
- B.** Nominal
- C.** Ordinal
- D.** Ratio

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A research firm surveys hospital nurses about burnout. The survey collects names and employee IDs so the firm can follow up, but promises that no individual answers will be shared with hospital managers. The study is funded by a nursing association and runs for six weeks. How should the firm describe its privacy promise?

- A.** Anonymous
- B.** Individual answers can go to managers, since they employ the nurses
- C.** No consent needed, because an association funds it
- D.** Confidential, not anonymous

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A coffee brand wants to know how its new cold brew is perceived. The team reads a published industry report on beverage trends and also runs its own taste test with 200 shoppers. ~~The brand's warehouse is in Seattle, and it just hired a new CMO.~~ Which is primary data?

Target word: primary data source **Decides:** Collected firsthand for this question.

A	Correct	Gathered by the brand for this study.
B	Adjacent category	Only the taste test was collected firsthand.
C	Inverted operation	Existing data from another source: secondary.
D	Absolute language	Experiments and observation are primary too.

2. Answer A Concept 2

A bank wants a survey of its 50,000 customers that reflects the exact mix of its three customer tiers. It divides customers by tier and randomly selects from each tier in proportion to its size. ~~The survey will be sent by email on a Tuesday.~~ Which sampling method is this?

Target word: sampling method **Decides:** Random draws within each subgroup.

A	Correct	Random selection inside defined strata.
B	Dropped element	Ignores the division into tiers.
C	Confusion set	Cluster picks whole groups at random.
D	Adjacent category	Quota fills groups non-randomly.

3. Answer C Concept 2 · transfer

A researcher stops shoppers at one mall entrance until she has 50 men and 50 women. What kind of sample is this?

Target word: classify the sample **Decides:** Subgroup counts filled without random selection.

A	Inverted operation	Not everyone had an equal chance of selection.
B	Confusion set	No random selection within groups.
C	Correct	Targets counts, picks whoever is available.
D	Absolute language	Covering groups does not make selection random.

4. Answer B Concept 3

A retailer wants to estimate the share of customers who would join a loyalty program within ± 5 percentage points at 95% confidence ($z = 1.96$), with no prior estimate, so $p = 0.5$. ~~The retailer has 120,000 customers and operates in four states.~~ What minimum sample size is needed?

Target word: minimum n **Decides:** $1.96^2 \times 0.25 \div 0.05^2 = 384.16$.

A	Dropped element	Forgets to square z.
B	Correct	Round 384.16 up.
C	Inverted operation	Rounds down, falling short of the target.
D	Camouflage	Takes 10% of the customer base; population size barely matters here.

5. Answer A Concept 3 · transfer

Keeping 95% confidence and $p = 0.5$, a researcher cuts the margin of error from ± 5 to ± 2.5 points. What happens to the required sample size?

Target word: effect of halving E **Decides:** E is squared in the denominator.

A	Correct	$1.96^2 \times 0.25 \div 0.025^2 = 1,536.6$.
B	Inverted operation	More precision needs more data.
C	Absolute language	Margin of error drives n directly.
D	Dropped element	Forgets that E is squared.

6. Answer B Concept 4

An A/B test compares a new checkout page with the old one. H_0 is no difference in conversion. The test gives $p = 0.03$ with $\alpha = 0.05$. ~~The new page uses a green button, and the test ran for 14 days.~~ What should the analyst conclude?

Target word: decision **Decides:** $p (0.03) \leq \alpha (0.05)$.

A	Inverted operation	That applies when $p > \alpha$.
B	Correct	p is below α .
C	Camouflage	The test compares whole pages, not one feature.
D	Confusion set	Misreads the p-value as the probability of H_1 .

7. Answer D Concept 4 · transfer

With $\alpha = 0.01$, a test returns $p = 0.04$. What is the conclusion?

Target word: decision at a stricter α **Decides:** $0.04 > 0.01$.

A	Wrong stem number	Uses 0.05 instead of the stated 0.01.
B	Inverted operation	p would need to be 0.01 or less.
C	Absolute language	Failing to reject does not prove H_0 .
D	Correct	p exceeds α .

8. Answer B Concept 5

A company concludes that a new ad raised sales, when in reality the ad had no effect. ~~The ad starred a famous chef and cost \$2 million.~~ What error is this?

Target word: error type **Decides:** Claimed an effect that was not real.

A	Adjacent category	Random variation in a sample, not a decision error.
B	Correct	Rejected a true null.
C	Inverted operation	Type II misses a real effect.
D	Confusion set	Bias from who does not answer a survey.

9. Answer D Concept 5 · transfer

A test fails to detect a real effect of a price change on demand. Which error is this, and what reduces its chance?

Target word: error type and fix **Decides:** Missed a real effect.

A	Confusion set	A stricter α raises the chance of Type II.
B	Inverted operation	Wrong error type.
C	Absolute language	Low power can hide real effects.
D	Correct	More data makes real effects easier to detect.

10. Answer C Concept 6

A survey for a 24-hour gym chain ~~with 60 locations~~ asks: 'How satisfied are you with our friendly staff and clean facilities?' ~~The survey will go to 10,000 members by text.~~ What is the main problem?

Target word: question flaw **Decides:** Two topics in one question, with loaded wording.

A	Absolute language	Loaded wording biases answers.
B	Camouflage	10,000 is plenty; sample size is not the flaw.
C	Correct	Answers cannot be separated and are nudged positive.
D	Adjacent category	The scale type is not the problem.

11. Answer C Concept 7

A survey asks runners to rank five shoe brands from 1 (favorite) to 5 (least favorite). ~~Responses came from 800 runners at a marathon expo.~~ What level of measurement is this?

Target word: measurement level **Decides:** Order without equal gaps.

A	Confusion set	Assumes equal distances between ranks.
B	Dropped element	Ignores the ranking.
C	Correct	Ranks show order only.
D	Adjacent category	Needs equal gaps and a true zero.

12. Answer D Concept 8

A research firm surveys hospital nurses about burnout. The survey collects names and employee IDs so the firm can follow up, but promises that no individual answers will be shared with hospital managers. ~~The study is funded by a nursing association and runs for six weeks.~~ How should the firm describe its privacy promise?

Target word: privacy label **Decides:** Identities are collected but protected.

A	Confusion set	Names and IDs are collected, so it is not anonymous.
B	Inverted operation	Breaks the confidentiality promise.
C	Camouflage	Funding source does not remove consent duties.
D	Correct	The firm knows who answered but will not disclose it.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Primary vs. secondary data	Primary data is collected firsthand for the current question; secondary data already exists from another source.	Calling a published report primary because it is new to you.	Who collected the data and for what purpose.
2	Probability vs. non-probability sampling	Stratified: random draws within defined subgroups; cluster: randomly pick whole groups; quota: fill subgroup counts non-randomly; convenience: whoever is handy.	Calling quota sampling stratified.	Whether selection within groups is random.
3	Sample size for a proportion	$n = z^2 \times p(1 - p) \div E^2$, rounded up; halving the margin of error quadruples n.	Forgetting to square z or E, or rounding down.	A confidence level, a margin of error and p.
4	p-value decision rule	Reject H_0 when $p \leq \alpha$; otherwise fail to reject (never 'accept' H_0).	Comparing p to the wrong α , or reading $1 - p$ as the chance H_1 is true.	A p-value and an α level.
5	Type I vs. Type II errors	Type I = false positive (reject a true H_0); Type II = false negative (miss a real effect); more power from a larger sample cuts Type II.	Swapping the two, or lowering α to fix a Type II error.	An effect that was claimed but not real, or real but missed.
6	Survey question bias	Avoid double-barreled questions (two topics) and leading wording that assumes the answer.	Blaming sample size for a bad question.	A question containing 'and' plus flattering words.
7	Levels of measurement	Nominal = labels; ordinal = ranked order; interval = equal gaps, no true zero; ratio = equal gaps and a true zero.	Treating rankings as interval data.	Rank, rating or count.
8	Research ethics: confidential vs. anonymous	Anonymous means the researcher cannot link answers to a person; confidential means identity is known but protected.	Calling identifiable data anonymous.	Names or IDs collected with a promise of privacy.

Course scope (public description): Market Research provides learners with a knowledge of the role of marketing research in strategic decision-making. Marketers need to systematically collect and analyze data to develop insights and make decisions. Learners demonstrate proficiency in the fundamentals of market research by practicing statistical methods for analyzing and acting on market data. This course teaches sampling techniques, survey development, data analysis, ethical considerations, and hypothesis testing. Learners will apply their knowledge using a market research simulation. Emphasis is placed on the interpretation and use of results to demonstrate how to communicate information. In this course, learners will also develop awareness for continually monitoring emerging and competitive trends given the dynamic digital marketing landscape.